



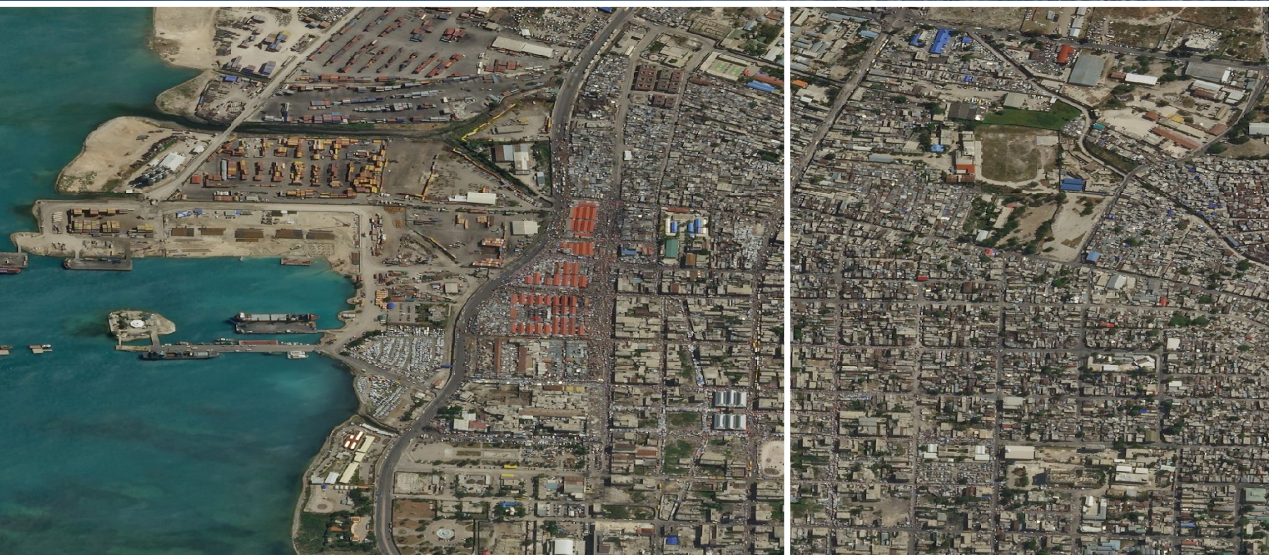
HAITI LANDCOVER CHANGE POST EARTHQUAKE

David Kimmel



INTRO

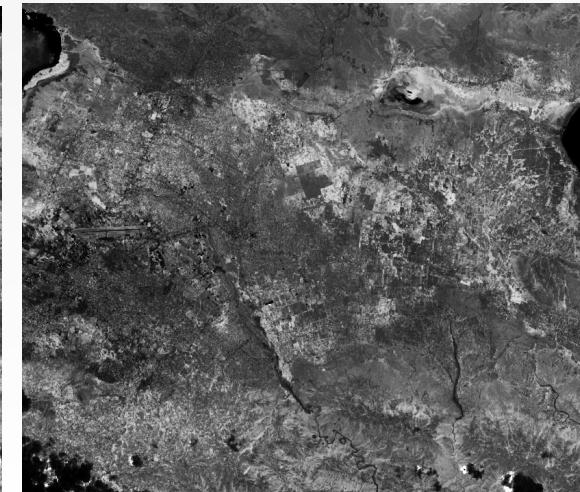
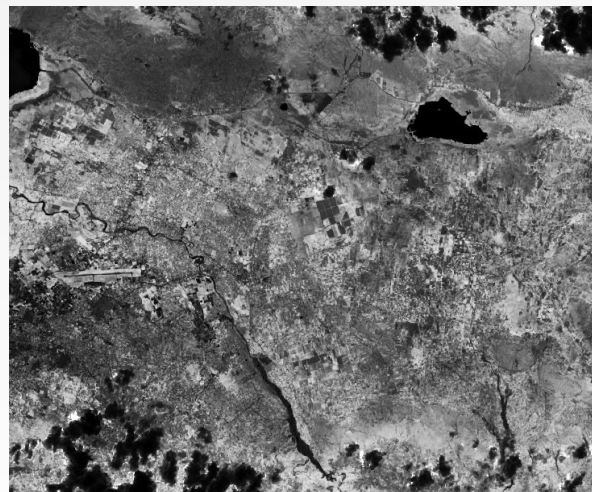
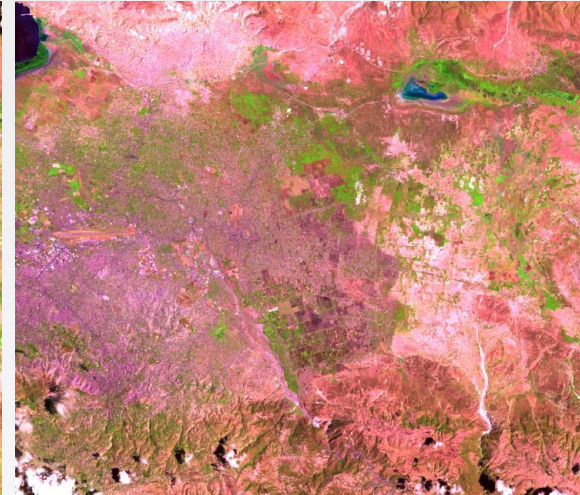
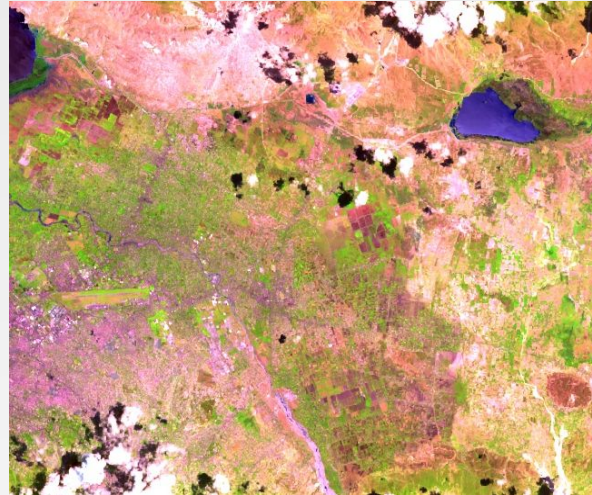
- ▶ What does landcover change in Haiti reveal about urban sprawl and vegetation loss in the years following the 2010 earthquake
- ▶ Landsat 8 imagery from 2013-2021
- ▶ Sentinel 2 imagery from 2016-2023
- ▶ Analyze NIR reflectance and Classify land cover
- ▶ Hypothesis: Urban sprawl post earthquake has increased and caused vegetation loss around Port-au-Prince Haiti



METHODOLOGY

- Using GEE to create a study area and time period from 2013-2021
- Two images depict early and late in SWIR-NIR-R
- NDVI is created on both images

```
5 // *****
6 // Block 1
7 var earlyImg_year = 2013; // Year of the early image
8 var earlyImg_startDay = '05-01'; // Date of the window to start looking for decent image
9 var earlyImg_endDay = '06-30'; // End of the window to look for images.
10
11 var lateImg_year = 2021; // Year of the late image
12 var lateImg_startDay = '04-01'; // Date of the window to start looking for decent image
13 var lateImg_endDay = '05-30'; // End of the window to look for images.
14
15 // *****
16 // Block 2
17 // Load in a geometry and find the centroid. Use the centroid to
18 // identify the appropriate Landsat images.
19
20 var studyarea = ee.FeatureCollection(geometry) // load in the geometry. for part 3 of the lab, you'll do your own geometry
21 var centroid = studyarea.geometry().centroid(3) // use "1" just as a minimum error distance
22 var bands_to_use = ['B2', 'B3', 'B4', 'B5', 'B6', 'B7']
23
24 // This will convert the image data into surface reflectance units.
25 var scaleTdata = function(img){
26   return ((img.multiply(0.0000275)).add(-0.2)).multiply(10000).toInt16();
27 };
28
29
30 var earlyImgCollection = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2') //SR
31   .filterDate(earlyImg_year+'-'+earlyImg_startDay, earlyImg_year+'-'+earlyImg_endDay)
32   .filterBounds(centroid) //finds only images that overlap this point.
33   .select(['SR_B2', 'SR_B3', 'SR_B4', 'SR_B5', 'SR_B6', 'SR_B7'], ['B2', 'B3', 'B4', 'B5', 'B6', 'B7'])
34   .map(scaleTdata)
35   .select(bands_to_use);
36
37 var earlyImg = ee.Image(earlyImgCollection.sort('CLOUD_COVER').first()).clip(studyarea)
38
39
40 var lateImgCollection = ee.ImageCollection('LANDSAT/LC08/C02/T1_L2') //SR
41   .filterDate(lateImg_year+'-'+lateImg_startDay, lateImg_year+'-'+lateImg_endDay)
42   .filterBounds(centroid) //finds only images that overlap this point.
43   .select(['SR_B2', 'SR_B3', 'SR_B4', 'SR_B5', 'SR_B6', 'SR_B7'], ['B2', 'B3', 'B4', 'B5', 'B6', 'B7'])
44   .map(scaleTdata)
45   .select(bands_to_use);
46
47 var lateImg = ee.Image(lateImgCollection.sort('CLOUD_COVER').first()).clip(studyarea)
48
49
50 // add these to the map, constrained to the geometry, using
51 // visualization parameters
52 var viz = {bands:['B6','B5','B4'], min:[0, 400,0], max:[3500, 5000,3000]}
53
54 Map.addLayer(earlyImg, viz, 'Early image')
55 Map.addLayer(lateImg, viz, 'Late image')
56 Map.centerObject(centroid, 9); // change the zoom factor if you want.
57
58 // *****
59 // Block 3
60 //
61 // Calculate NDVI for both images, subtract
62 var earlyNDVI = earlyImg.normalizedDifference(['B5', 'B4']).multiply(1000).toInt(); // multiply by 1000 to make it to rescale
63 var lateNDVI = lateImg.normalizedDifference(['B5', 'B4']).multiply(1000).toInt();
64
65 var diffBand = lateNDVI.subtract(earlyNDVI).clip(studyarea);
66
67 var diffViz = {min:-1000, max:1000};
68 var BvViz = {min:0,max:1000};
69
70 Map.addLayer(earlyNDVI, BvViz, 'Early NDVI');
71 Map.addLayer(lateNDVI, BvViz, 'Late NDVI');
72 Map.addLayer(diffBand, diffViz, 'Late minus early NDVI')
73
74 // // // *****
```

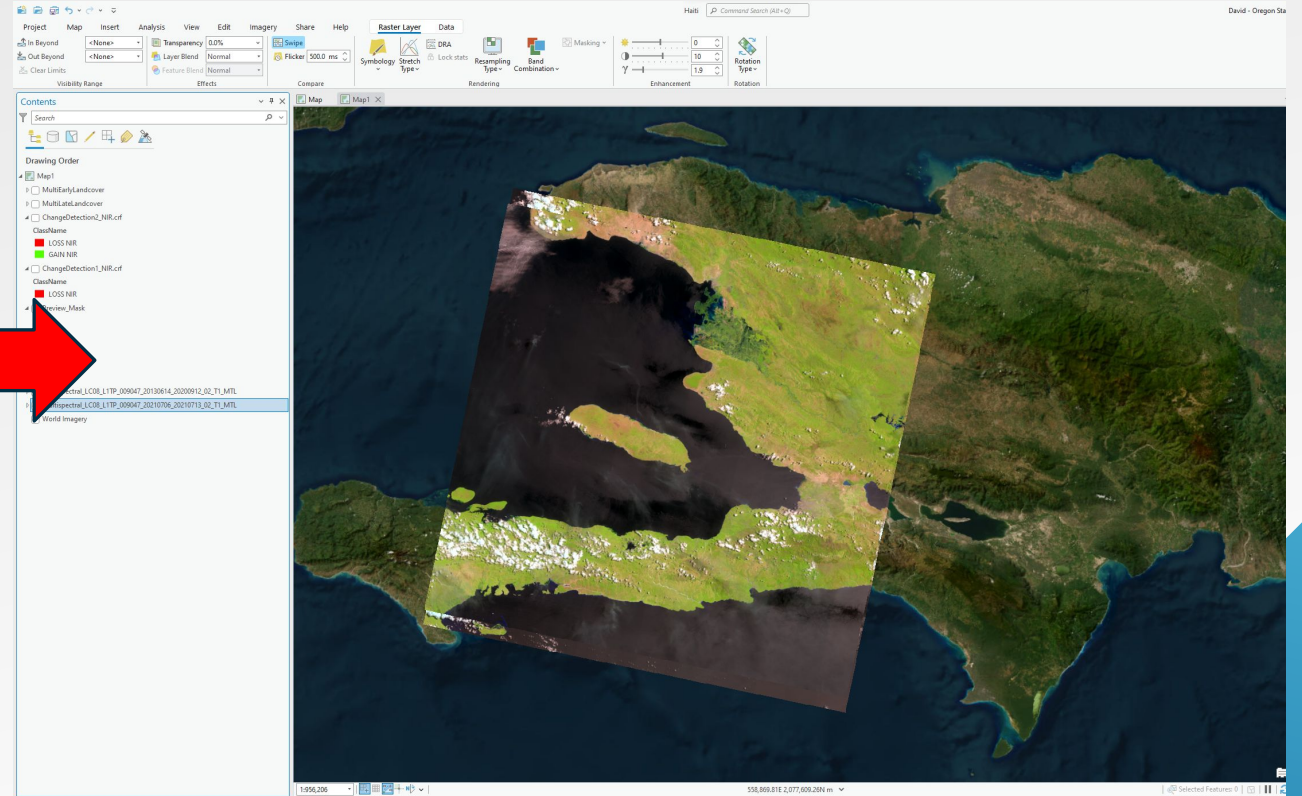
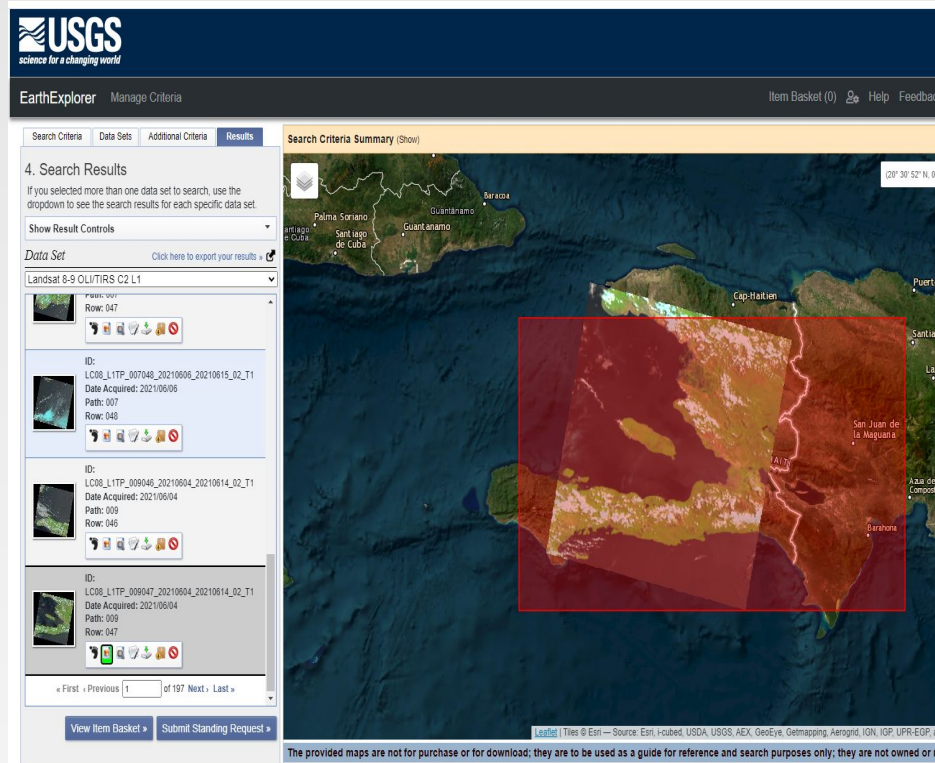


METHODOLOGY

- Late NDVI is subtracted from the Early NDVI
- Problems with further analysis using GEE. Unable to further exploit imagery with limited coding experience



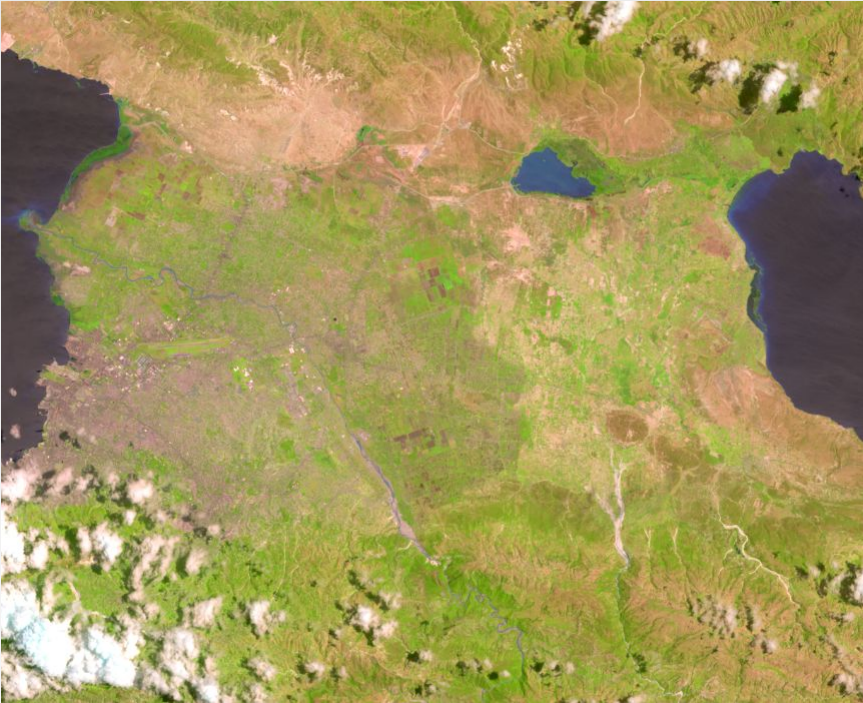
METHODOLOGY



Landsat 8 Imagery was collected from the USGS Earth Explorer website and imported into ArcPRO

METHODOLOGY

2013



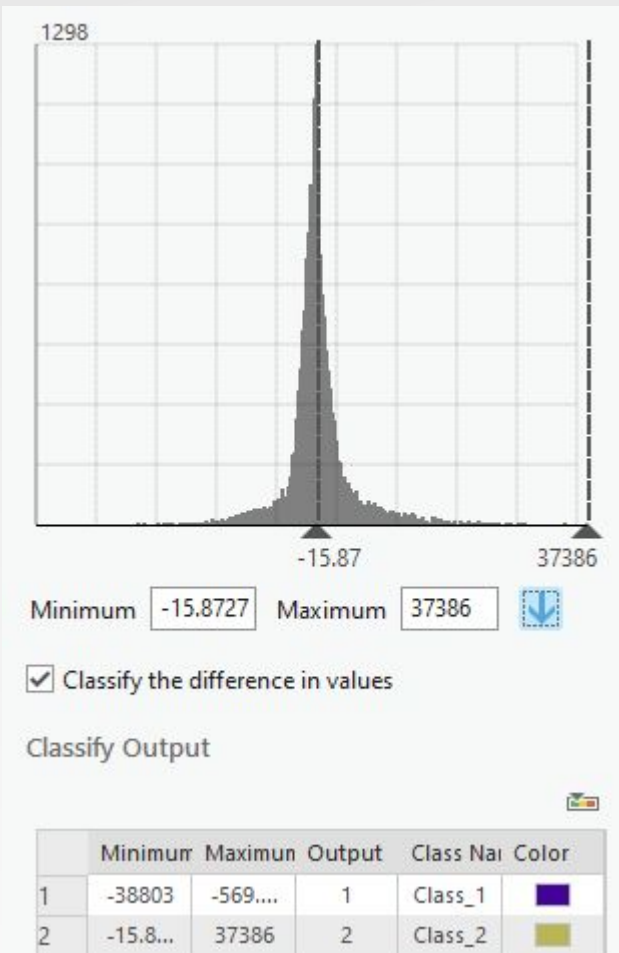
2021



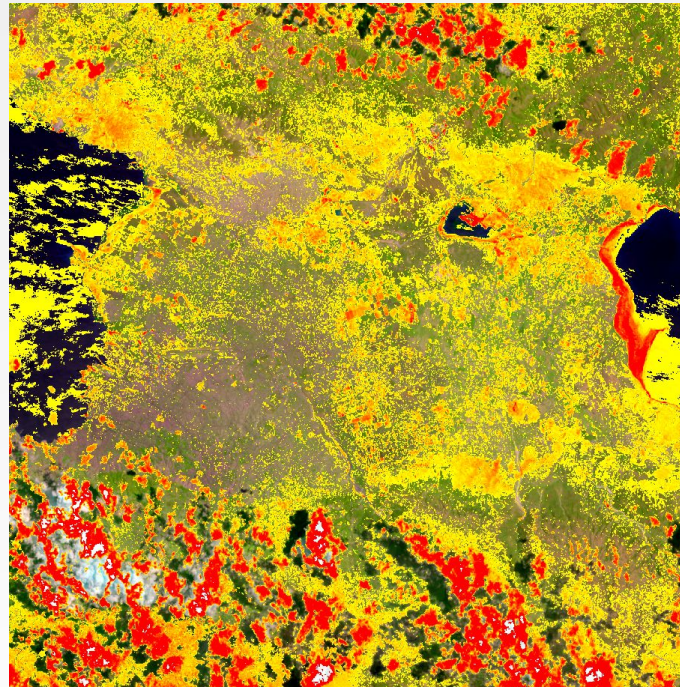
- Imagery was collected and imported into ArcPRO displayed by bands 6,5,4 (SWIR,NIR,Red)
- Original imagery was clipped to the study area of Port-au-Prince

METHODOLOGY

NIR Change Detection

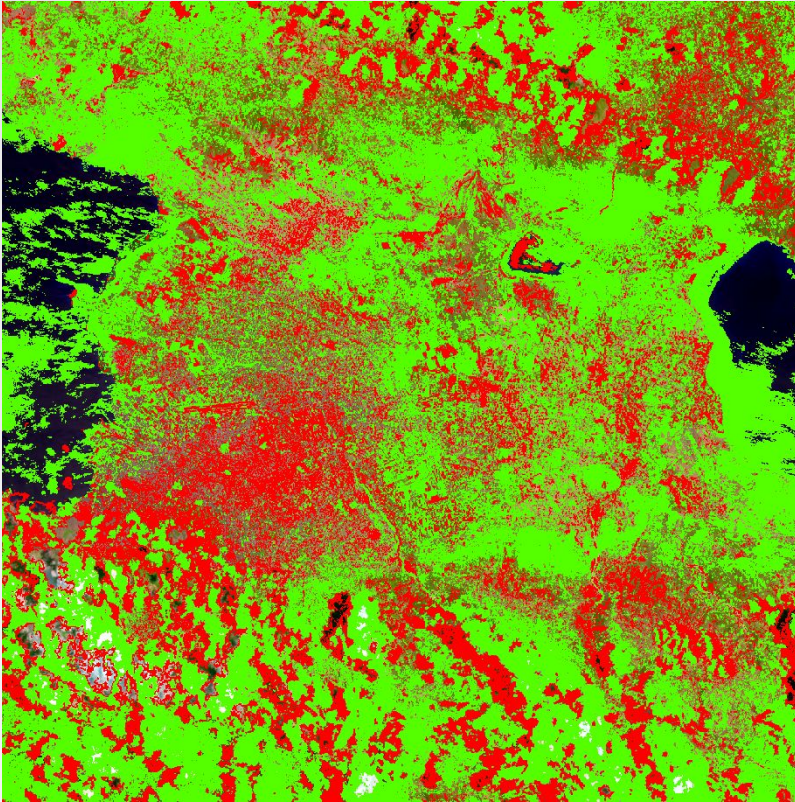


- Using ArcPRO change detection wizard, I analyze the change in the NIR band from the first image to the second.
- This is the absolute pixel value change in Band 5 from 2013 to 2021 images
- Using the histogram I created two categories, NIR change below 0 and above 0, indicating increase and decrease in NIR



METHODOLOGY

NIR Change Detection



- This process produced a map indicating areas that lost vegetation in red and areas that increased vegetation in green.
- At this point my hypothesis is either wrong or needs further exploration. There is an equal amount of gain and loss

METHODOLOGY

Creating a Land Cover map in ArcPRO

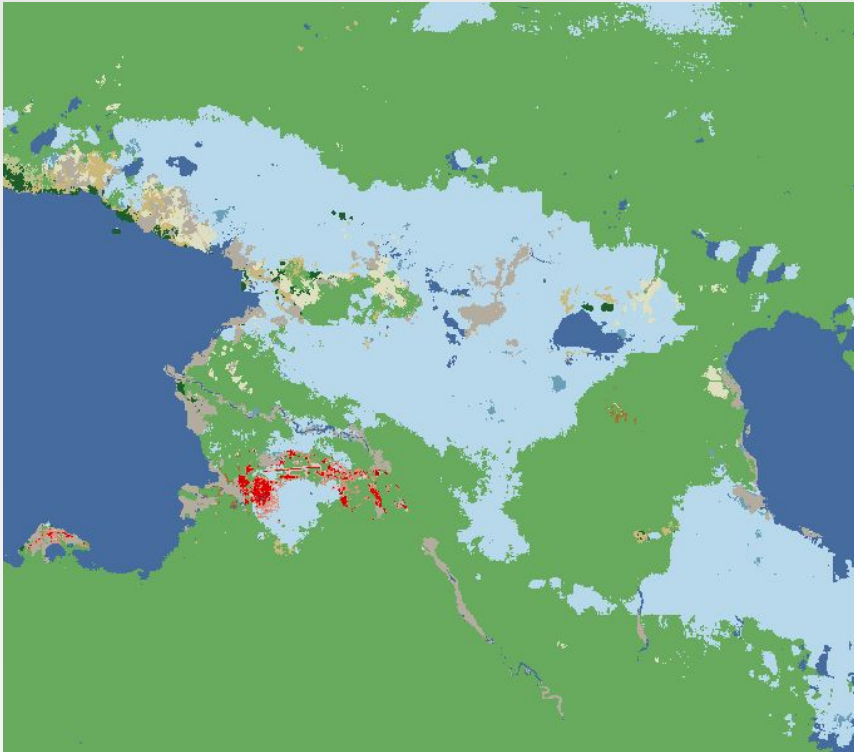
- I will use a Deep learning model to perform land cover classification on the Landsat 8 imagery. This model is acquired from ESRI
- This model uses the [UNet](#) model architecture implemented in ArcGIS API for Python
- This model has an overall accuracy of 77 percent.
- This model has been trained on the National Land Cover Database (NLCD) 2016 with the same Landsat 8 scenes that were used to produce the database.

The table below summarizes the precision, recall and F1-score of the model on the validation dataset.

Class	Collection 2 Level 2 Imagery			Collection 1 Level 1 Imagery		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Open Water	0.96	0.97	0.96	0.95	0.97	0.96
Perennial Snow/Ice	0.86	0.69	0.77	0.49	0.94	0.64
Developed, Open Space	0.51	0.38	0.44	0.43	0.38	0.4
Developed, Low Intensity	0.52	0.46	0.49	0.47	0.48	0.47
Developed, Medium Intensity	0.54	0.5	0.52	0.49	0.54	0.51
Developed, High Intensity	0.67	0.54	0.6	0.55	0.68	0.61
Barren Land	0.76	0.59	0.66	0.6	0.77	0.68
Deciduous Forest	0.74	0.81	0.78	0.78	0.76	0.77
Evergreen Forest	0.77	0.82	0.79	0.8	0.82	0.81
Mixed Forest	0.56	0.47	0.51	0.5	0.53	0.51
Shrub/Scrub	0.82	0.82	0.82	0.84	0.81	0.83
Herbaceous	0.78	0.74	0.76	0.79	0.77	0.78
Hay/Pasture	0.7	0.74	0.72	0.67	0.75	0.71
Cultivated Crops	0.87	0.91	0.89	0.91	0.9	0.9
Woody Wetlands	0.7	0.68	0.69	0.67	0.68	0.68
Emergent Herbaceous Wetlands	0.72	0.54	0.62	0.54	0.61	0.57

METHODOLOGY

2013



2021



Class

- Open Water
- Perennial Snow/Ice
- Developed, Open Space
- Developed, Low Intensity
- Developed, Medium Intensity
- Developed, High Intensity
- Barren Land
- Deciduous Forest
- Evergreen Forest
- Mixed Forest
- Shrub/Scrub
- Herbaceous
- Hay/Pasture
- Cultivated Crops
- Woody Wetlands
- Emergent Herbaceous Wetlands

- Landcover maps indicate that there has been more vegetation growth than loss during the study period.

METHODOLOGY

2016



2021



- Sentinel 2 imagery acquired for the study area from Copernicus Open Access Hub
- Imagery is clipped to the study area

METHODOLOGY

2016



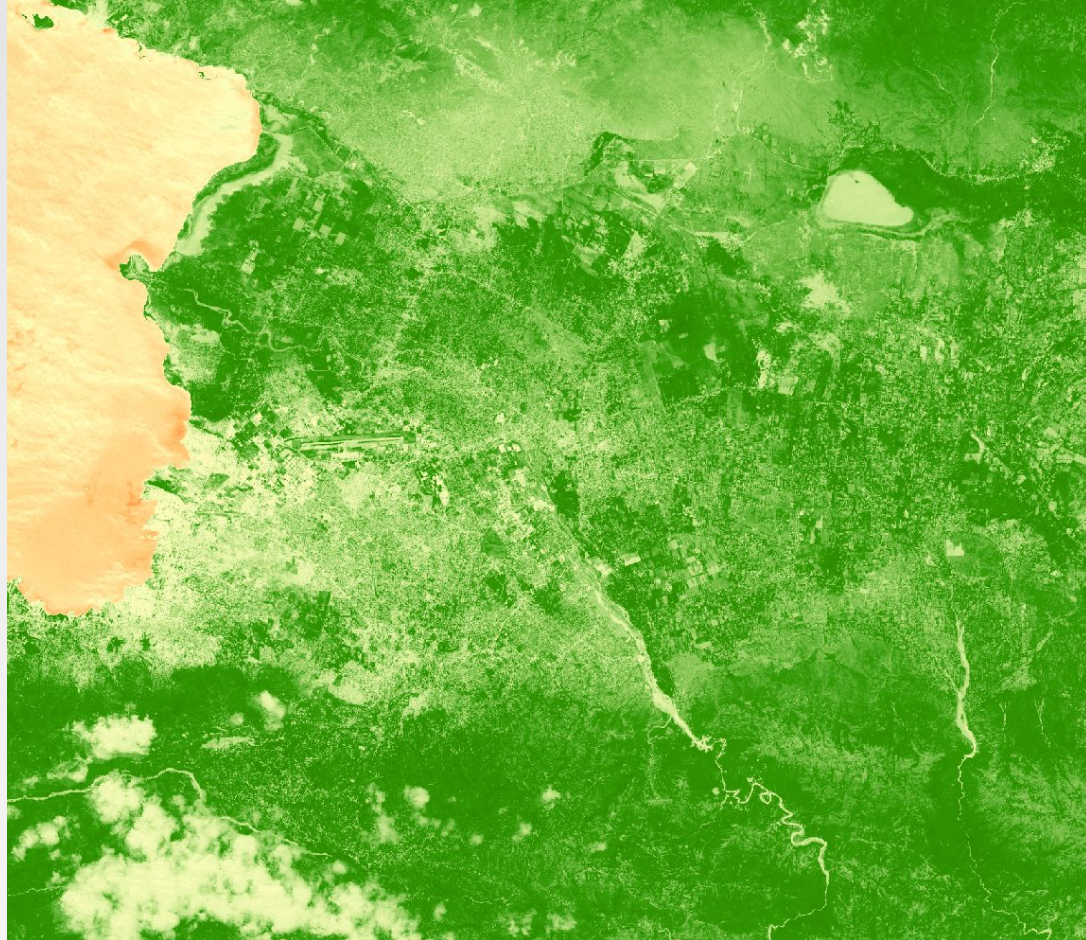
2021



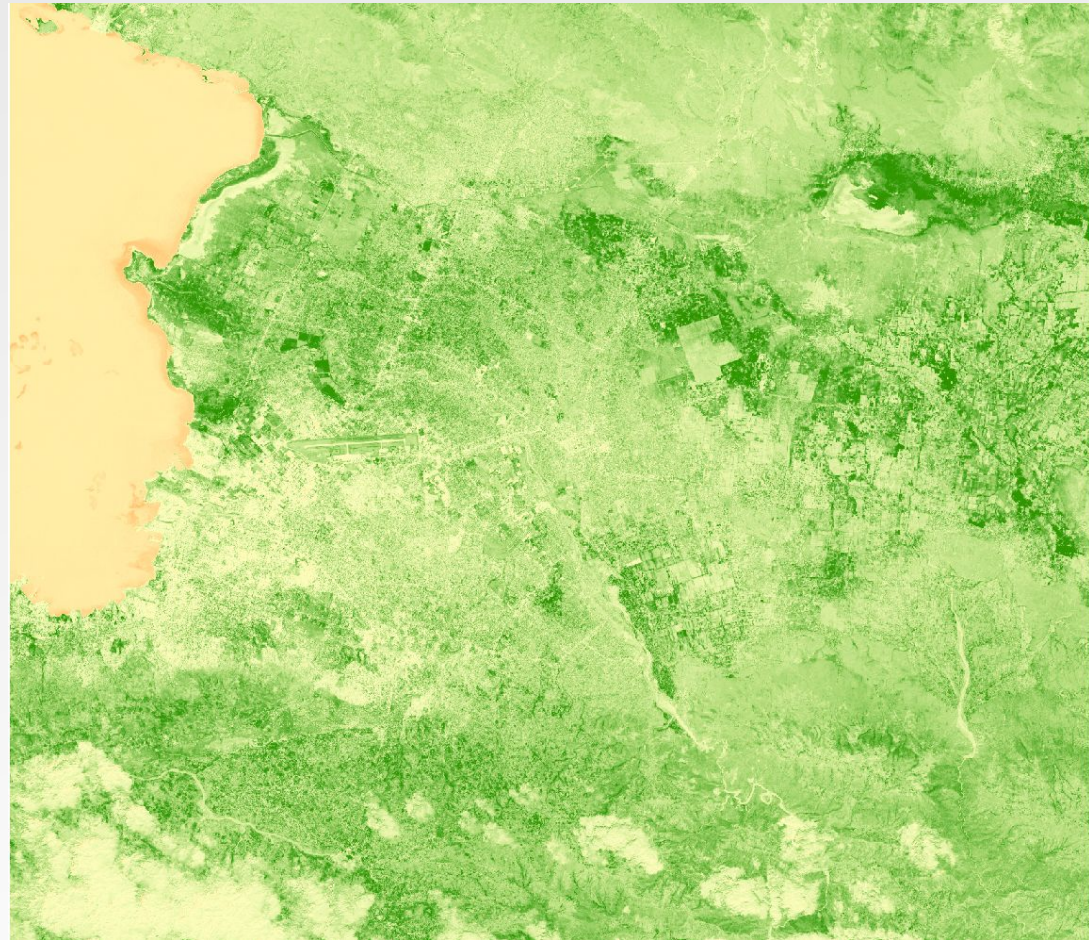
- We can clearly see areas like this that have lost vegetation to Urban Sprawl in the Sentinel Imagery from 2016 to 2023

METHODOLOGY

2016

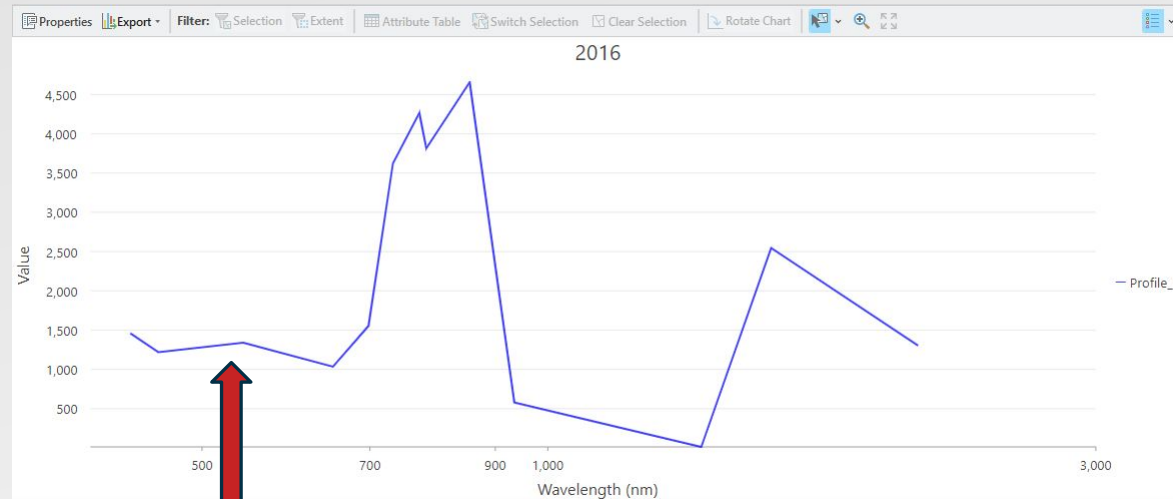


2021



- Colored NDVI was created to show the loss in vegetation between 2016 and 2023
- Obvious signs of an overall decrease in vegetation correlating with urban sprawl

METHODOLOGY



- Loss of vegetation in several cells indicating